*Delivering Military Advantage through multi-national geospatial interoperability*

# DGIWG-NWP-LIGHTNING-001

**Edition 1.0**

**Dated 16 July 2026**

# DGIWG NWP Earth Observations Lightning Profile

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# I. ABSTRACT

This document defines the DGIWG NWP Earth Observations Lightning Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

# II. KEYWORDS

The following are keywords to be used by search engines and document catalogues.

cube, radius, items, datetime, frequency_of_lightning_flashes_per_unit_area, lightning flashes, lightning

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# III. PREFACE

This document was prepared by Met Office, DGIWG.

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# IV. SECURITY CONSIDERATIONS

No security considerations have been made for this document.

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# V. SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to DGIWG:

- Met Office
- DGIWG

# VI. SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

**Table** — Submitters

| NAME | AFFILIATION |
|---|---|
| Austria | Institute of Military Geography |
| France | Institut National de l'Information Géographique et Forestière (IGN) |
| Finland | Finnish Defence Forces |
| Germany | Bundeswehr Geoinformation Centre (BGIC) |
| NATO | NATO Maritime GEOMETOC Centre of Excellence |
| United Kingdom | Met Office (MO), UK Strategic Command, DSTL |
| United States | National Geospatial-Intelligence Agency (NGA) |

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# 1. SCOPE

This standard defines the DGIWG NWP Earth Observations Lightning Profile. It specifies requirements and conformance tests for implementations of this profile.

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# 2. CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.dgiwg.org/std/edr/1.0/conf/nwp_earth_observations_lightning

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# 3. NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

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# 4. TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

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# 5. REQUIREMENTS

**NOTE:** Conformance with this profile is assessed against its requirements ("shall") only. Recommendations ("should") and permissions ("may") are provided as informative guidance: they are not conformance-tested and an implementation is not made non-conformant by not following them. See the quality-assessment summary emitted by the profile tooling for a count of the recommendations and permissions defined by this profile.

| REQUIREMENTS CLASS 1 | |
|---|---|
| **IDENTIFIER** | `http://www.dgiwg.org/std/edr/1.0/req/nwp_earth_observations_lightning` |
| **CONFORMANCE CLASS** | Conformance class A.1: `http://www.dgiwg.org/std/edr/1.0/conf/nwp_earth_observations_lightning` |
| **TARGET-TYPE** | DGIWG NWP Earth Observations Lightning Profile Profile Standard |
| **NORMATIVE STATEMENTS** | Requirement 1: `/req/nwp_earth_observations_lightning/items-endpoint`<br>Requirement 2: `/req/nwp_earth_observations_lightning/cube-endpoint`<br>Requirement 3: `/req/nwp_earth_observations_lightning/radius-endpoint` |

| REQUIREMENT 1 | |
|---|---|
| **IDENTIFIER** | `/req/nwp_earth_observations_lightning/items-endpoint` |
| **INCLUDED IN** | Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_earth_observations_lightning` |
| **STATEMENT** | The service SHALL provide a /collections/earth_observations_lightning/items endpoint. |
| **A** | The service SHALL return GeoJSON FeatureCollection. |
| **B** | Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property. |
| **C** | Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z. |

| REQUIREMENT 2 | |
|---|---|
| **IDENTIFIER** | `/req/nwp_earth_observations_lightning/cube-endpoint` |
| **INCLUDED IN** | Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_earth_observations_lightning` |

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| REQUIREMENT 2 | |
|---|---|
| **STATEMENT** | The service SHALL provide a /collections/earth_observations_lightning/cube endpoint. |
| **A** | The service SHALL return CoverageJSON FeatureCollection. |
| **B** | Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property. |
| **C** | Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z. |

| REQUIREMENT 3 | |
|---|---|
| **IDENTIFIER** | `/req/nwp_earth_observations_lightning/radius-endpoint` |
| **INCLUDED IN** | Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_earth_observations_lightning` |
| **STATEMENT** | The service SHALL provide a /collections/earth_observations_lightning/radius endpoint. |
| **A** | The service SHALL return CoverageJSON FeatureCollection. |
| **B** | Each feature SHALL include frequency_of_lightning_flashes_per_unit_area property. |
| **C** | Each feature SHALL include datetime property in the format YYYY-MM-DDTHH:MM:SSZ e.g. 2026-04-12T21:20:50.52Z. |

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# ANNEX A
# (NORMATIVE)
# ABSTRACT TEST SUITE

| CONFORMANCE CLASS A.1 | |
|---|---|
| **IDENTIFIER** | `http://www.dgiwg.org/std/edr/1.0/conf/nwp_earth_observations_lightning` |
| **REQUIREMENTS CLASS** | Requirements class 1: `http://www.dgiwg.org/std/edr/1.0/req/nwp_earth_observations_lightning` |
| **CONFORMANCE TESTS** | Abstract test A.1-1: `/conf/nwp_earth_observations_lightning/items-endpoint`<br>Abstract test A.1-2: `/conf/nwp_earth_observations_lightning/cube-endpoint`<br>Abstract test A.1-3: `/conf/nwp_earth_observations_lightning/radius-endpoint` |

**Example 1**

| | |
|---|---|
| label | /conf/nwp_earth_observations_lightning/items-endpoint |
| subject | /req/nwp_earth_observations_lightning/items-endpoint |
| test-purpose | Validate that items endpoint is correctly implemented. |

Send GET request to /collections/earth_observations_lightning/items.

Verify response Content-Type is application/json.

Verify each feature contains frequency_of_lightning_flashes_per_unit_area property.

Verify each feature contains datetime property.

**Example 2**

| | |
|---|---|
| label | /conf/nwp_earth_observations_lightning/cube-endpoint |
| subject | /req/nwp_earth_observations_lightning/cube-endpoint |
| test-purpose | Validate that cube endpoint is correctly implemented. |

Send GET request to /collections/earth_observations_lightning/cube with a valid coordinate in crs_4326 and a radius of 1km.

Verify response Content-Type is application/json.

Verify each feature contains frequency_of_lightning_flashes_per_unit_area property.

Verify each feature contains datetime property.

Verify each feature is in projection crs_4326.

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**Example 3**

| | |
|---|---|
| label | /conf/nwp_earth_observations_lightning/radius-endpoint |
| subject | /req/nwp_earth_observations_lightning/radius-endpoint |
| test-purpose | Validate that radius endpoint is correctly implemented. |

Send GET request to /collections/earth_observations_lightning/radius with a valid coordinate in crs_4326.

Verify response Content-Type is application/json.

Verify each feature contains frequency_of_lightning_flashes_per_unit_area property.

Verify each feature contains datetime property.

Verify each feature is in projection crs_4326.

Verify each feature is returned within the radius area.

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